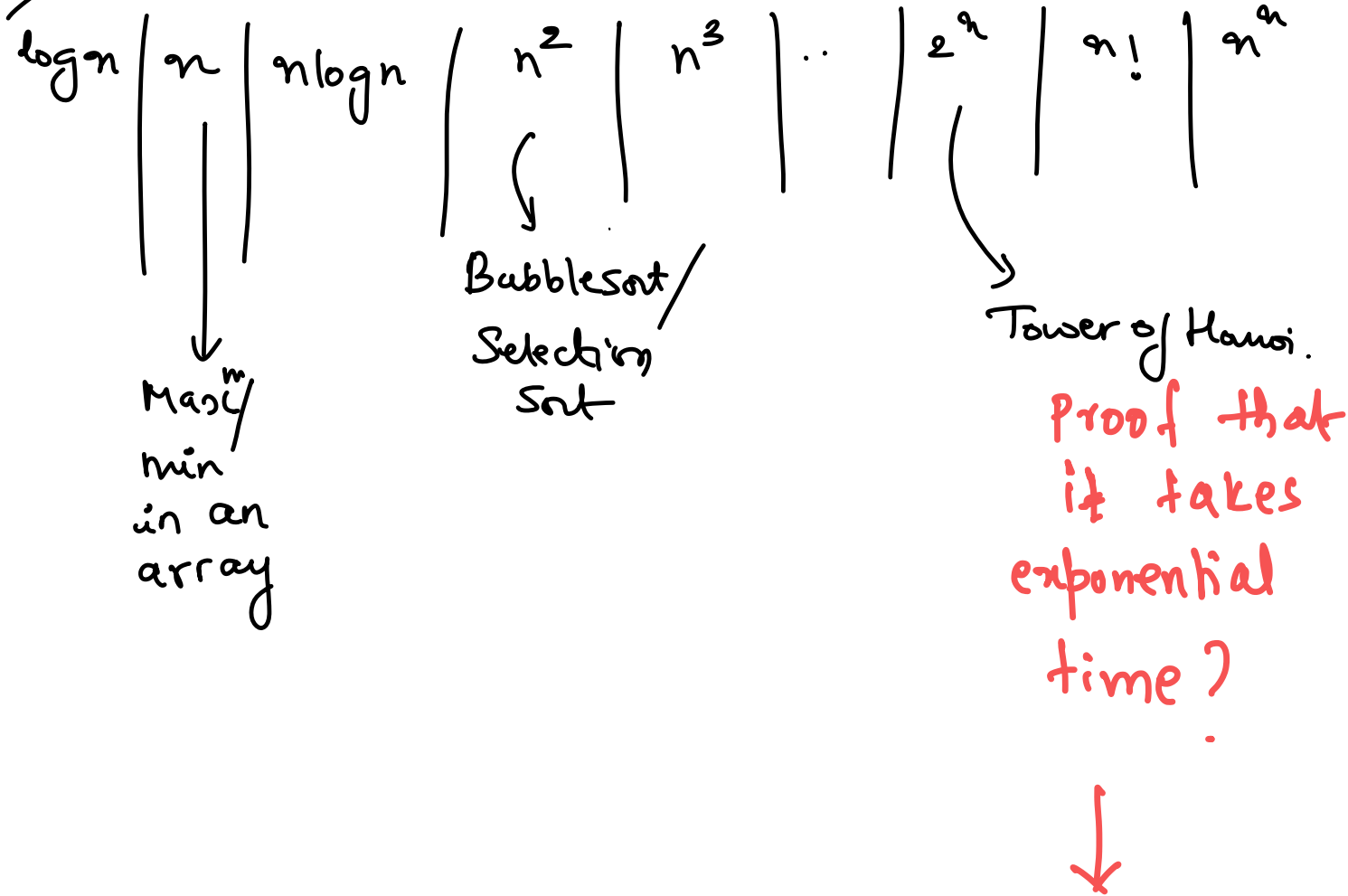
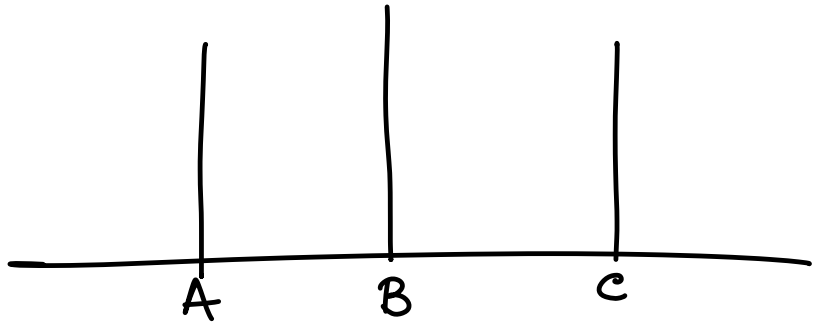
11/8/26


$\log n$
n
$n \log n$
n^2
n^3
..
2^n
$n!$
n^n
Bubblesort/
Selection
Sort
Tower of Hanoi.
Max^m/
min
in an
array
Proof that
it takes
exponential
time?

Tower of Hanoi :

Consists of 3 pegs, n disks.



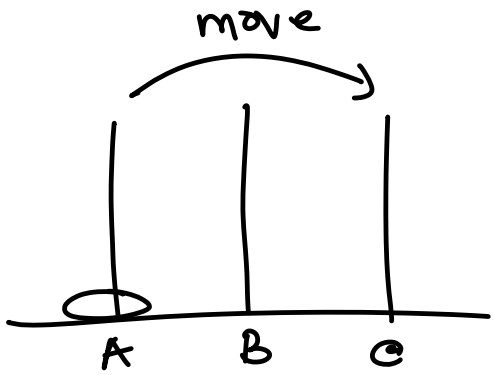
Rules:

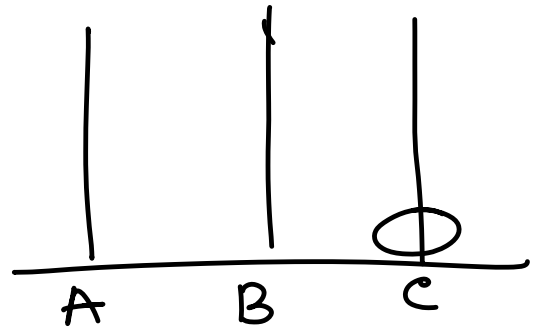
① Move one disk at a time

② Do not put larger disk onto a smaller one (No bullying).

Q: Can you move all disks from Tower A to Tower B?

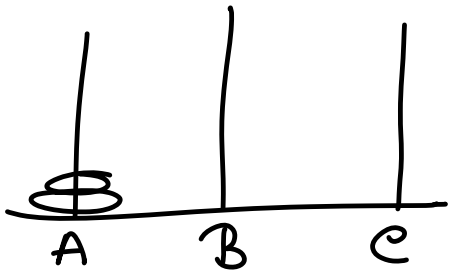
$n=1$:



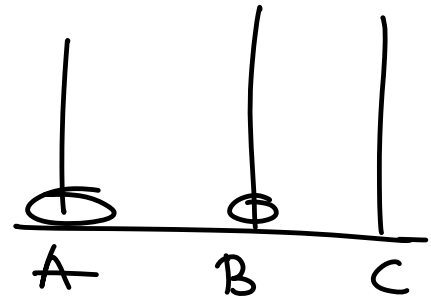


No of steps = 1.

$n=2$:

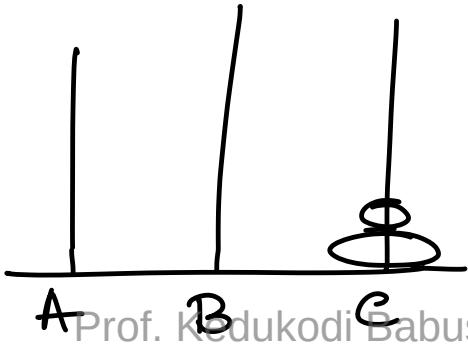


Step 1 →

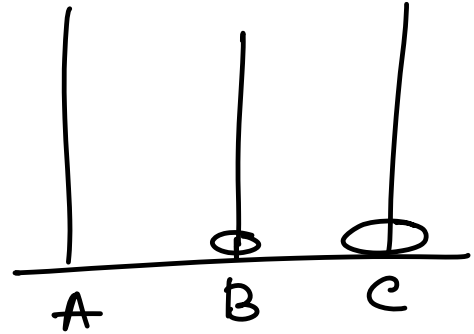


No of steps = 3.

↓ Step 2



← Step 3



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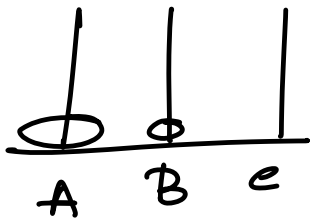
Q: Can we do it with less than 3 steps?

As there are n disks, n moves are required.

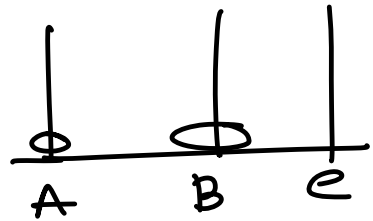
(Take n = 2.)

Q: Can we do it in 2 moves?

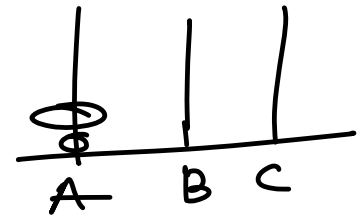
If Yes, Situations of initial State.



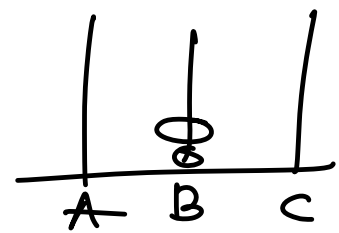
OR



OR



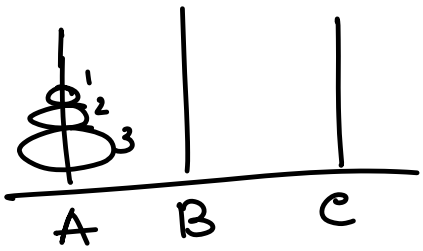
OR



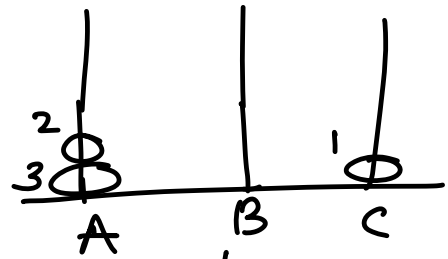
None of these are allowed as initial states.

Hence 3 steps is optimum number of moves for $n = 2$ disks.

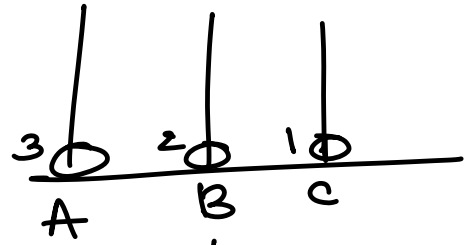
n=3.



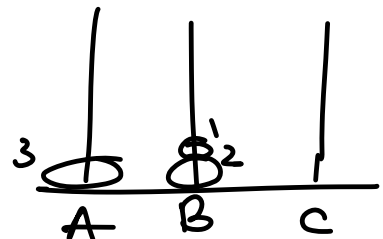
S.1



S.2

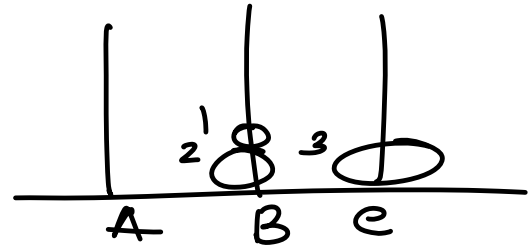


S.3

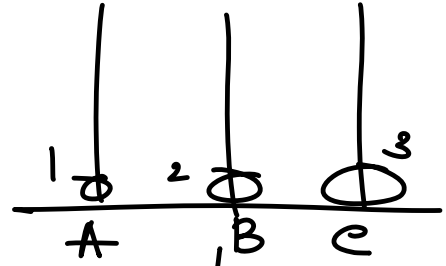


S.4

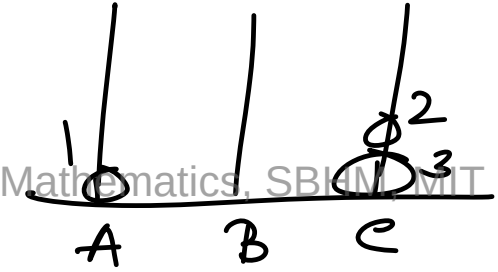
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5.5

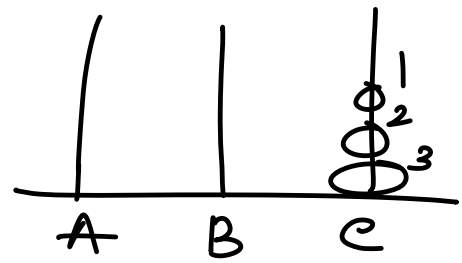


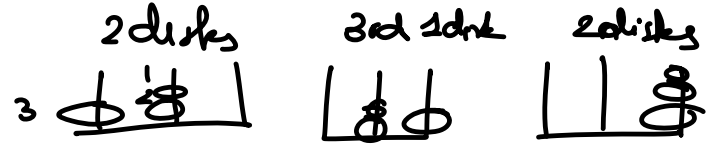
5.6



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5.7.





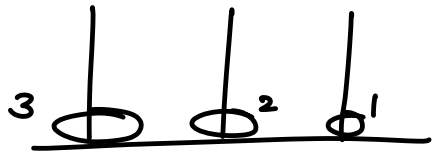
$3 + 1 + 3$

No of steps = 7

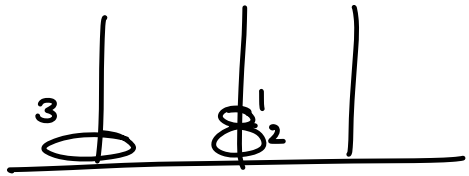
Step 1:



Step 2:

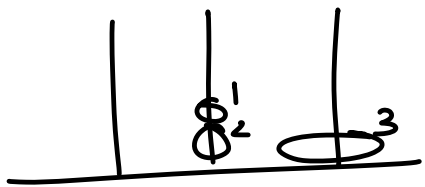


Step 3:



$T(n-1)$

Step 4:

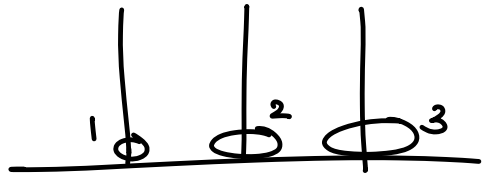


(largest disk will be moved exactly once)

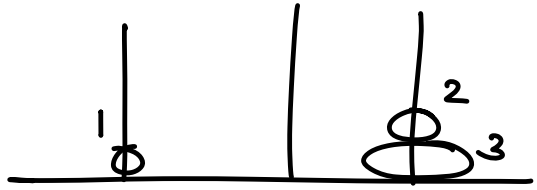
(Middle step)

1

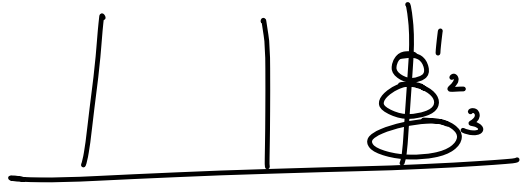
Step 5:



Step 6:



Step 7.



$T(n-1)$

no of disks	Min no of moves	$T(n)$
1	1	
2	3	
3	7	
⋮		
n	~~2^n-1~~	Good guess: 2^n-1.

2^n-1 (proved).

(Symmetric)

$$T(n) = T(n-1) + 1 + T(n-1) \quad \text{with } T(1) = 1$$

$$= 2T(n-1) + 1$$

↳ Recurrence for Tower of Hanoi problem.

Solving the recurrence:

$$T(n) = 2T(n-1) + 1$$

$$= 2\{2T(n-2) + 1\} + 1$$

$$= 2^2 T(n-2) + 2 + 1$$

$$= 2^2\{2T(n-3) + 1\} + 2 + 1$$

$$= 2^3 T(n-3) + 2^2 + 2 + 1$$

.
.
.

$$= 2^{n-1} T\left(n-(n-1)\right) + 2^{n-2} + 2^{n-3} + \cdots 2+1$$

$$= 2^{n-1} T\left(n-n+1\right) +$$

$$= 2^{n-1} T(1) + 2^{n-2} + 2^{n-3} + \cdots + 2+1$$

$$= 2^{n-1} + 2^{n-2} + 2^{n-3} + \cdots + 2+1$$

$$= \frac{2^n - 1}{2-1}$$

(Geometric Series)

com. ratio = 2.

first term = 1

$$= 2^n - 1$$

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$2^n - 1 \in \Theta(2^n)$.

(use limit).

	speed of computer per second.	Time to solve Tower of Hanio is solved.
$n=100$ disks	10^{12} (disks) operations per second.	million years.

Finish $2^{100} - 1 \simeq 2^{100}$ operations.

$1 \text{ sec} \quad 10^{12}$

$? \quad 2^{100}$

$$\frac{2^{100}}{10^{12}} = \frac{1.29 \times 10^{30}}{10^{12}}$$

$$= 1.29 \times 10^{18} \text{ seconds.}$$

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1 sm 60s.
? x

$$= \frac{1.29 \times 10^{18}}{60} \text{ min}$$

$$= 2.1 \times 10^{16} \text{ min}$$

$$= \frac{2.1 \times 10^{16}}{60} \text{ hours}$$

$$= 35 \times 10^{14} \text{ hours}$$

$$= \frac{3.5 \times 10^{14}}{24} \text{ days}$$

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$$= 1.4 \times 10^{13} \text{ days}$$

$$= \frac{1.4 \times 10^{13}}{365} \text{ years}$$

$$= 4.0. \times 10^{10} \text{ years.}$$

Age of Earth $= 4.5 \times 10^{9}$ years.

(4.5 billion years)

Note:

* Exponential algorithms are useful in cryptography, and encryption.
* For other purposes, Exponential algorithms may be useful only for small cases.

Q. on Sorting:

